Validating the jNO PEEC solver against analytical solutions

Three codes, three answers, and which one the mathematics supports

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Three solvers disagree on the same power module

The measurement. Loop inductance between the DC+ and DC– terminals of one layout, at 1 MHz, on geometry all three codes read from the same file.

	loop inductance	
Ansys Q3D (`ACLoop`)	20.641 nH	reference
`peec_coarse`	20.5 nH	unattributed
pypeec 5.8.0	21.4 nH	not converged
jNO	26.4 nH	+28 %

Why it is not obvious who is wrong. No reference here is authoritative: two are numbers without a method attached, and the third is still moving under mesh refinement.

So test against something with a closed-form answer

The case. One straight copper bar, 40 × 4 × 0.51 mm, driven end to end at 1 kHz — the current fills the section, so both the resistance and the inductance are known.

	R	error	L	error
analytic	338.0 $\mu\Omega$	—	27.207 nH[1]	—
jNO	337.2 $\mu\Omega$	−0.2 %	27.731 nH	+1.9 %
pypeec	332.7 $\mu\Omega$	−1.6 %	26.673 nH	−2.0 %

The result. jNO lands on the exact DC resistance and sits inside Grover's own accuracy. pypeec approaches from below and is still 2 % short at its finest affordable grid.

[1] Grover, *Inductance Calculations*, 1946

The reference was biased, but jNO still saturates

What that corrects. pypeec under-estimates when under-converged, and its module numbers rose monotonically the whole way, 51.3 → 53.6 nH. Most of the module gap was the reference.

What survives. jNO's inductance stops moving above 1 MHz — 26.448, 26.435, 26.430 nH at 1, 3, 10 MHz. A resolved solve would keep falling, so the mesh has stopped capturing current crowding rather than the physics having stopped.

Open. What Q3D's `ACLoop` actually reports. Its mesh is 3416 tetrahedra with a 3.4 mm RMS edge, against a 66 μm skin depth.

Corrected, the two PEEC codes agree

jNO, as run	26.05 nH
quantised plane gap — one z-cell forces 0.44 mm where the stack has 0.37 surface impedance knows the skin depth, the partial inductance does not	−0.73 −2.34
jNO, corrected pypeec, converged	22.97 nH 22.5–23.0 nH
Ansys Q3D (`ACLoop`)	20.641 nH

Priced and dismissed. Frequency (jNO is flat 1–10 MHz), capacitance (663× the inductive impedance), bond wires (–5 % at 56× the real cross-section).

Where that leaves it. Two independent PEEC codes now agree. Ansys sits ~10 % below both, so the open question is what `ACLoop` reports — not why jNO is high.